

EXPLORATORY DATA ANALYSIS

USING VISUALIZATION

ASSIGNMENT

Titanic Passenger Analysis Assignment Objective The objective of this assignment is to perform Exploratory Data Analysis (EDA) on the Titanic dataset using Python libraries such as Pandas, Matplotlib, and Seaborn. The analysis focuses on passenger distribution, survival patterns, passenger class, fare analysis, and relationships among variables.

Google Colab Notebook Code # Import required libraries import pandas as pd import numpy as np import matplotlib.pyplot as plt import seaborn as sns # Load dataset # Upload Titanic.csv file in Google Colab before running df = pd.read_csv('Titanic.csv') # Display first 5 rows print(df.head())

Basic Data Exploration # Dataset information print(df.info()) # Statistical summary print(df.describe()) # Check missing values print(df.isnull().sum())

Handling Missing Values # Fill missing Age values with median if 'Age' in df.columns: df['Age'].fillna(df['Age'].median(), inplace=True) # Fill missing Embarked values with mode if 'Embarked' in df.columns: df['Embarked'].fillna(df['Embarked'].mode()[0], inplace=True) # Drop Cabin column if too many missing values if 'Cabin' in df.columns: df.drop('Cabin', axis=1, inplace=True) # Check again print(df.isnull().sum())

Question 1: Create a countplot to show the number of passengers by gender.

Answer: Countplot to Show Number of Passengers by Gender

```
plt.figure(figsize=(6,5))
sns.countplot(x='Sex', data=df, palette='Set2')
plt.title('Number of Passengers by Gender')
plt.xlabel('Gender')
plt.ylabel('Passenger Count')
plt.show()
```

Interpretation

- The countplot shows the distribution of male and female passengers.

- Male passengers were higher in number compared to female passengers.
- This indicates that most travelers on the Titanic were men.

Question 2: Create a bar chart to show the number of passengers in each passenger class using different colors.

Answer: Bar chart for Number of Passengers in Each Passenger Class

```
plt.figure(figsize=(7,5))
sns.countplot(x='Pclass', data=df, palette='viridis')
plt.title('Passengers in Each Passenger Class')
plt.xlabel('Passenger Class')
plt.ylabel('Count')
plt.show()
```

Interpretation

- Passenger Class 3 had the highest number of passengers.
- Passenger Class 1 had fewer passengers compared to Class 3.
- This suggests that most passengers traveled in lower-class accommodations.

Question 3: Create a histogram to show the distribution of passenger age.

Answer: Histogram for Distribution of Passenger Age

```
plt.figure(figsize=(8,5))
plt.hist(df['Age'], bins=20)
plt.title('Age Distribution of Passengers')
plt.xlabel('Age')
plt.ylabel('Frequency')
plt.show()
```

Interpretation

- Most passengers were between 20 and 40 years old.
- Very few elderly passengers were present.
- The histogram shows that the Titanic mainly carried young adults.

Question 4: Create a boxplot to detect outliers in the fare column.

Answer: Boxplot to Detect Outliers in Fare Column

```
plt.figure(figsize=(8,5))  
  
sns.boxplot(x=df['Fare'], palette='coolwarm')  
  
plt.title('Boxplot of Fare') plt.xlabel('Fare')  
  
plt.show()
```

Interpretation

- The boxplot shows several outliers in the Fare column.
- Some passengers paid extremely high ticket fares.
- Most passengers paid lower ticket prices.

Question 5: Create a scatterplot to

- A. Show the relationship between age and fare.**
- B. Analyze the relationship between family size (SibSp + Parch) and fare.**

Answer: (a). Scatterplot Between Age and Fare

```
plt.figure(figsize=(8,5))  
  
sns.scatterplot(x='Age', y='Fare', data=df)  
  
plt.title('Relationship Between Age and Fare')  
  
plt.xlabel('Age') plt.ylabel('Fare')  
  
plt.show()
```

Interpretation

- There is no strong relationship between age and fare.
- Passengers of all age groups paid different ticket prices.
- Higher fares were mainly related to passenger class rather than age.

(b). Scatterplot Between Family Size and Fare

```
# Create FamilySize column  
if 'SibSp' in df.columns and 'Parch' in df.columns:  
  
    df['FamilySize'] = df['SibSp'] + df['Parch']  
  
  
plt.figure(figsize=(8,5))  
  
sns.scatterplot(x='FamilySize', y='Fare', data=df)
```

```
plt.title('Relationship Between Family Size and Fare')  
plt.xlabel('Family Size')  
plt.ylabel('Fare')  
plt.show()
```

Interpretation

- Families with larger sizes sometimes paid higher fares.
- Small families and solo travelers were more common.
- Fare values varied significantly across family sizes.

Question 6: Create a countplot to show survival count by gender.

Answer: Countplot to Show Survival Count by Gender

```
plt.figure(figsize=(7,5)) sns.countplot(x='Sex', hue='Survived', data=df, palette='Set1')  
plt.title('Survival Count by Gender')  
plt.xlabel('Gender')  
plt.ylabel('Count')  
plt.legend(title='Survived')  
plt.show()
```

Interpretation

- Female passengers had a higher survival rate.
- Most male passengers did not survive.
- This reflects the “women and children first” rescue policy.

Question 7: Create a bar chart to show survival by passenger class.

Answer: Bar Chart to Show Survival by Passenger Class

```
plt.figure(figsize=(7,5))  
sns.barplot(x='Pclass', y='Survived', data=df, palette='magma')  
plt.title('Survival Rate by Passenger Class')  
plt.xlabel('Passenger Class')  
plt.ylabel('Average Survival Rate')  
plt.show()
```

Interpretation

- First-class passengers had the highest survival rate.
- Third-class passengers had the lowest survival rate.
- Wealthier passengers had better access to safety measures.

Question 8: Create a heatmap to show correlation between numerical variables.

Answer: Heatmap Showing Correlation Between Numerical Variables

```
plt.figure(figsize=(10,7))

correlation = df.corr(numeric_only=True)

sns.heatmap(correlation, annot=True, cmap='coolwarm')

plt.title('Correlation Heatmap')

plt.show()
```

Interpretation

- Survival has a positive relationship with Fare.
- Passenger class negatively correlates with survival.
- Some variables have weak relationships with each other.

Question 9: Identify which passenger class paid the highest fare on average.

Answer: Identify Which Passenger Class Paid the Highest Fare on Average

```
average_fare = df.groupby('Pclass')['Fare'].mean()

print(average_fare)
```

Interpretation

- First-class passengers paid the highest average fare.
- Third-class passengers paid the lowest average fare.
- Ticket prices depended strongly on passenger class.

Question 10: Write at least 5 insights from the dataset based on your visualizations.

Answer: Final Insights from the Dataset

Final Insights

1. Male passengers were more than female passengers on the Titanic.
2. Third-class passengers formed the majority of travelers.
3. Most passengers were young adults between 20–40 years.

4. Female passengers had a much higher survival rate compared to males.
5. First-class passengers had the best survival chances.
6. Higher ticket fares were mostly associated with first-class passengers.
7. The Fare column contained several outliers due to expensive tickets.
8. Passenger class strongly influenced both fare and survival probability.
9. Family size showed slight influence on fare values.
10. Survival was positively related to fare and negatively related to passenger class.

Conclusion

The Titanic dataset analysis helps in understanding passenger demographics, fare distribution, survival patterns, and class-based differences. Through various visualizations, it becomes clear that gender and passenger class played a major role in determining survival chances during the Titanic disaster.

Libraries Used

- Pandas
- NumPy
- Matplotlib
- Seaborn

Output

This assignment includes:

- Clean and readable Python code
- Visualizations with titles and labels
- Interpretation of each chart
- Final insights from the dataset